

Optimal signal tower placement by mixed integer linear programming

CDE2605 Final Report

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Abstract

This research presents a mixed-integer programming (MIP) framework based on the Generalized Assignment Problem (GAP) to address real-world allocation challenges. The primary application involves the optimal allocation of signal towers. The framework is then extended to simulate cost and pricing relationships for land reclamation projects, providing contractors with a quantitative tool to estimate optimal bidding prices and support profit-maximizing decisions. This research demonstrates the practical application of optimization in bridging operational planning with financial strategy, thereby guiding more sustainable project execution. Numerical experiments are also reported to validate the claims proposed in this research.

1 Introduction

In our increasingly interconnected world, telecom signals are always growing in demand, especially in metropolitan cities. To ensure the quality standard of signal coverage, we look into ways to fulfil the ever-growing need for signal in all areas, while keeping the number of signal towers at a minimum cost by selecting the appropriate tower type at suitable locations. In addition, we also consider the possibility of transforming previously unbuildable areas into viable tower placement sites, particularly if doing so could further minimise cost or meet the demand of areas that were previously unreachable. This can be achieved using optimisation software and techniques such as Gurobi and the branch-and-price algorithm, which are capable of handling large datasets efficiently.

Ultimately, this approach provides land reclamation contractors with an estimated price range to propose when negotiating with government agencies regarding the conversion of unbuildable land. This supports informed business decision-making aimed at maximising profit.

1.1 Problem Statement

The challenge lies in minimising the total cost of installing these towers while still meeting the communication fluctuating demand of all regions across the country. This problem requires a careful balance of cost efficiency and reliable service coverage within Singapore's unique urban landscape.

1.2 Objectives

Our alogirthm aim to minimise the total cost of installing telecom towers in Singapore while ensuring that the risk-sensitive communication demand across all regions in the country is adequately met. This involves determining the optimal locations, number and types of towers required to achieve effective coverage. This MIP model

seeks to balance cost efficiency with reliable connectivity to address Singapore's dense and diverse urban landscape, where some areas are previously unaccessible could be considered to be reclaimed given a certain cost. By addressing these considerations, our study aims to propose a cost-effective and scalable solution for future telecom infrastructure planning and relcamation vendors in densely populated environments.

2 Modelling the Problem

2.1 Assumptions and Data Collection

To model the problem effectively, several key assumptions are made based on available data and for ease of calculation.

2.1.1 Geographical Division

Singapore is divided into a grid of $500m \times 500m$ cells. This grid overlays the entire country, excluding offshore islands with negligible populations.

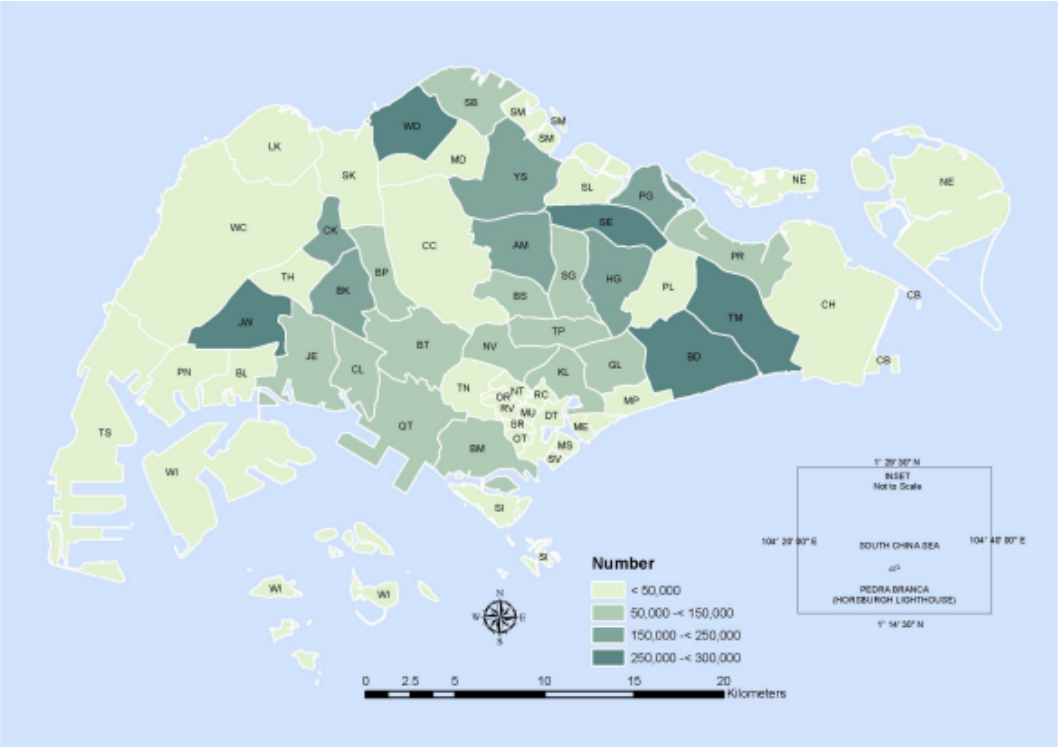


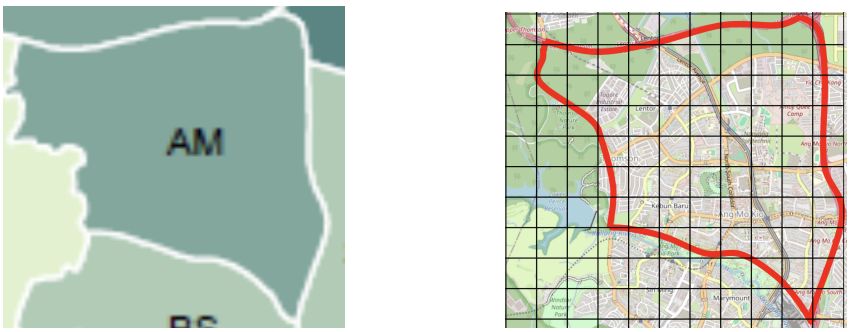
Figure 1: Resident Population by Planning Area, June 2025[1]

2.1.2 Population Distribution

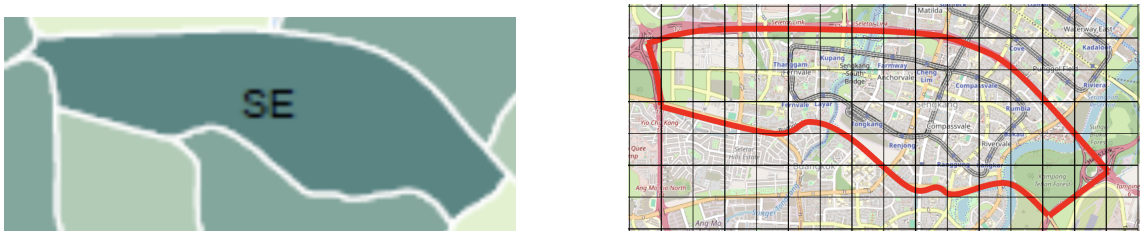
Resident population data as of June 2025[2] is used, sourced from the Singapore Department of Statistics. The population of each planning area is evenly distributed across the cells within that area due to the lack of more granular data, barring the

areas that are obviously not populated such as bodies of water and forested areas. Public areas such as roads and walkways are still considered as people still utilise these areas.

For example, Yishun has about 74 boxes encapsulated for a population of 229,520 for an average of about 3,100 per box, while Ang Mo Kio has about 46 boxes encapsulated for a population of 159,340 for an average of roughly 3,460 per box. Due to the development types within the Seletar area, there are only about 33 boxes encapsulated for a much smaller population of only 300 and an average of about 10 per box. Lastly, for Sengkang, there are about 44 boxes encapsulated for a population of 265,060 for an average of 6,020 per box. The examples are illustrated in Figure 2.



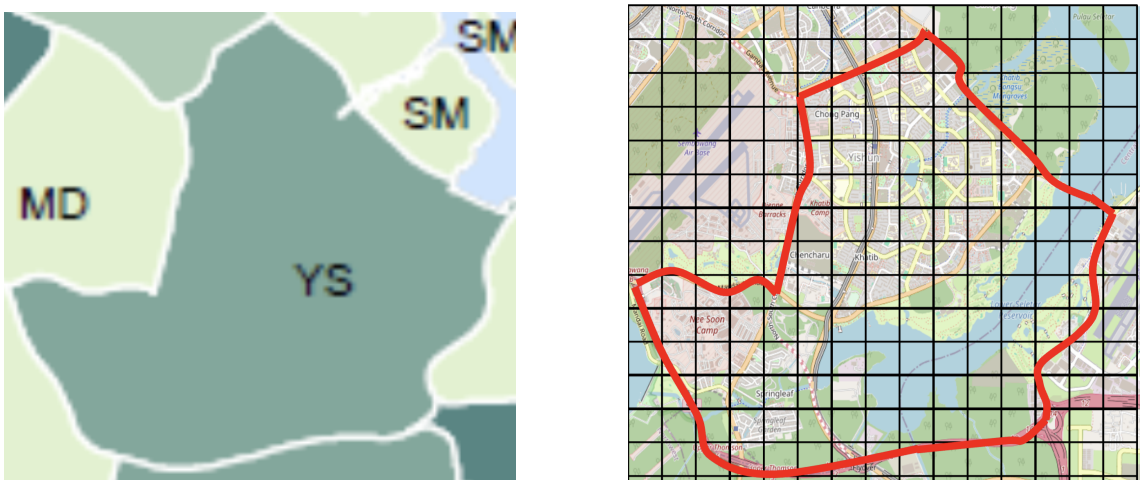
(a) Ang Mo Kio



(b) Sengkang



(c) Seletar



(d) Yishun

Figure 2: Example of Regions and Encapsulated Boxes

2.1.3 Initial Signal Range and Strength

Singapore uses Mobile Phone Base Stations with ranges of between 0.5km to 2km. To simplify this, we can assume each telecom tower has an effective signal range of up to 1.25km, covering its own cell, the immediate surrounding cells (8 cells), and the next layer of surrounding cells (16 cells), forming a coverage area of 25 cells, as visualised in Figure 3. Signal strength decreases with distance, following an inverse square law:

- **Central Cell**: Full strength (normalised to 1), serving up to 200 users.
- **Immediate Surrounding Cells**: Strength at 0.25 (since at double the distance, strength is $\frac{1}{2^2} = 0.25$), serving up to 50 users per cell.
- **Next Layer Surrounding Cells**: Strength at approximately 0.11 (since at triple the distance, strength is $\frac{1}{3^2} \approx 0.11$), serving up to 20 users per cell.

20	20	20	20	20
20	50	50	50	20
20	50	200	50	20
20	50	50	50	20
20	20	20	20	20

Figure 3: Simplified Signal Range and Strength in Grid

2.1.4 Initial Tower Capacity

Each tower can serve up to 920 users across its coverage area. This is based on current estimates where Singapore has approximately 6,750 towers serving 4.2 million residents, yielding an average of about 620 users per tower. For capacity planning, we assume a tower can handle up to 1,000 users, accounting for peak demand and future growth. This value was used in the rough calculation of the capacity per box area based on strength of the signal and then rounded. This then gave the values of 920 users at capacity.

Cell Type	Signal Distance	Signal Strength	Location Compatibility	Cost (rooftop estimate)
Micro	0 m	-30 to -40 dBm	Moderate sized outdoor and suburban areas	$50,000
	2 boxes	-70 to -90 dBm		$50,000
	3 boxes	-90 to -110 dBm		$50,000
Small	0 m	-30 to -50 dBm	Urban and dense areas	$20,000
	2 boxes	-60 to -80 dBm		$20,000
DAS	0 m	-20 to -30 dBm	Yes, for indoor, necessary when indoor coverage is critical	$5,000

Table 1: Overview of Cell Types [3][4]

2.2 Tower Types

We will be focusing on 3 types of cell towers, all of which are deployable on land and on buildings, at varying costs, as Singapore avoids using Macro cell towers;. The cell types are Micro cells, Small cells and the Distributed Antenna System (DAS).

The scoring system is created with the help of the Received Signal Strength Indicator categorisation system which categorises signal levels into different strength levels (each with their own descriptions). The RSSI categorisation varies based on the generations of technology standards for cellular networks ('G'), as shown in Table 2.

Generation / RSSI rating	**4G & 5G**
Excellent	≥ -65 dBm
Good	-65 to -75 dBm
Fair	-75 to -85 dBm
Poor	-85 to -95 dBm
No Signal	< -95 dBm

Table 2: RSSI Categorisation and Signal Values[5]

The signal strength for each RSSI rating is given in decibel milliwatts (dBm). Combining the estimated signal strengths and the scoring system (based on the RSSI rating standards), we arrived at our scores for each cell type at each distance from the source, as shown in Table 3.

Cell Type	Signal Distance	Singal Strength (dBm)	Location Compatibility	Compatible Networks	4G Score	5G Score
Micro	0 m 2 boxes 3 boxes	-30 to -40 -70 to -90 -90 to -110	Moderate sized outdoor and suburban areas	3G, 4G, 5G	4 1.333 1	4 1.33 1
Small	0 m 2 boxes	-30 to -50 -60 to -80	Urban and dense areas	3G, 4G, 5G	3.75 2	3.75 2
DAS	0 m	-20 to -30	Yes, for indoor, necessary when indoor coverage is critical	4G, 5G	4	4

Table 3: Summary of Cell Types and Their Corresponding Scores

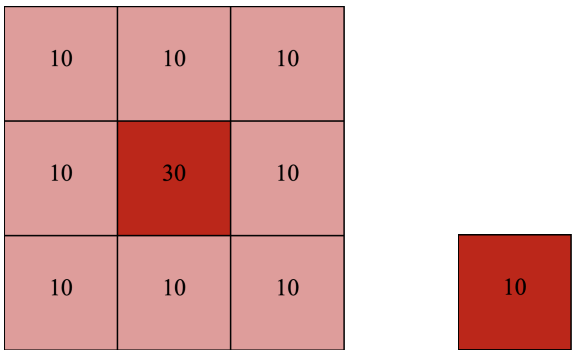
Another important property of the cell types is that they have different compatible networks, with the Micro cell being compatible with most generations of networks. At the same time, DAS is only compatible with 4G and 5G networks. Furthermore, due to the nature of the cell types, the signal distance they can maintain decreases from Micro cells to DAS, with Micro cells able to maintain signals for long distances for outdoor purposes while the DAS has a high quality but short signal range suitable for indoor purposes.

For the purposes of this report, we assume that the larger the score, the more people the signal can service adequately. We convert the scores arbitrarily into the number of people who will be satisfied with the service. This provides a simplified but effective way to check if the demand needs of each grid region have been met satisfactorily.

Cell Type	**User Capacity**	**Range**	**Cost**
Macro	Not considered, as Singapore avoids using large cell towers;		
Micro	$\leq 2,000$ **Central Cell Strength: 920** (We can place our initial estimate under this at 920)	$\leq 1,000$ m (We can use our initial estimate of 1.25 km where we allow the range to traverse 3 box widths out)	\$50,000
Small	≤ 100 **Central Cell Strength: 30** (We can place an arbitrary number of 110 as roughly an order of magnitude lower than the next highest cell type)	We can estimate this to be 2 box widths. Using a similar calculation, the central box can handle 30 with the surroundings handling 10.	\$20,000
DAS	≤ 30 **Central Cell Strength: 10** (We can place an arbitrary number of 10 as an order of magnitude lower than the next highest cell type)	We can estimate this to only supply to the box the tower is built on to handle 10	\$5,000

Table 4: Summary of Cell Types with User Capacity, Range, and Cost

Based on Table 4, we can develop the different ranges for the smaller towers as shown in Figure 4.



(a) (b)

Figure 4: Illustration of Ranges for the Smaller Towers

2.3 MIP Modelling

2.3.1 Decision Variables

Let X_{ij} be the number of towers to install in a cell (i, j). x_{ij} is a non-negative integer.

2.3.2 Parameters

- D_{kl}: Demand (number of users) in cell (k, l).
- c_{type}: Cost of installing telecom tower (based purely on type).
- $S_{mn,kl}$: Signal strength contribution from a tower at cell (m, n) to cell (k, l).
- C_{mn}: The set of cells covered by a tower at cell (m, n).

Note that (i, j), (k, l), (m, n) are equivalent and only written as such for clarity.

2.3.3 Constraints

Demand Satisfaction Constraints For every cell (k, l), the cumulative signal contributions from all towers covering the cell must meet or exceed the demand:

$$\sum_{(m,n)\in\mathcal{T}_{kl}} S_{mn,kl} \cdot x_{mn} \geq D_{kl}$$

where $\mathcal{T}_{kl}$ is the set of towers that can cover cell (k, l).

The maximum contribution to a central square from a tower at a coloured position and each of the x positions in each layer can be visualised in Figure 5. The signal is shaded for one tower for clarity. The largest tower type is used for reference, but this can be extrapolated to the other tower types as well.

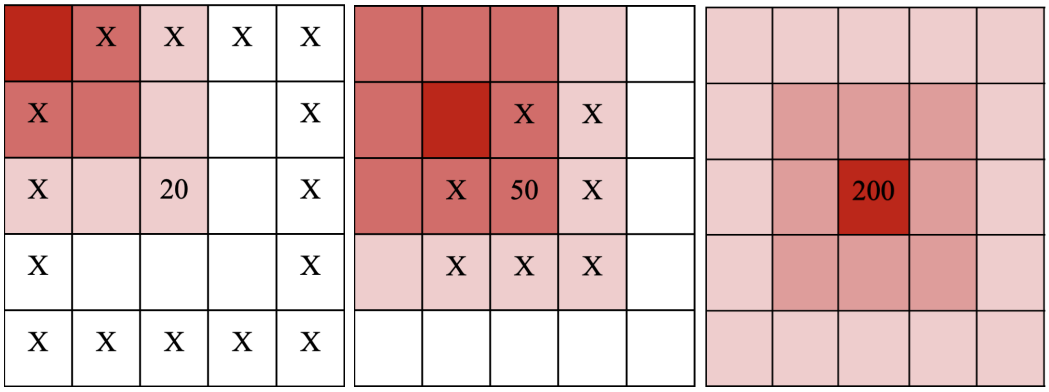


Figure 5: Visualisation of Maximum Signal Contribution from a Tower

Tower Capacity Constraints For each tower at cell (m, n), the total demand it serves cannot exceed its capacity which can be expressed as:

$$\sum_{(k,l)\in C_{mn}} S_{mn,kl} \cdot D_{kl} \leq \text{Capacity} \times x_{mn}$$

However, as every cell's demand can be served by multiple towers, this is embedded within the earlier demand satisfaction constraints where S_{mn} is the sum of all the respective max capacities providable from all towers in range of the area and does not have to be explicitly defined in the final LP problem.

Installation Constraints Cell towers cannot be built in certain areas such as water bodies, the sea, forested areas or public use areas unless paid a given reclamation cost.

Non-Negativity Constraints

$$X_{i,j} \geq 0$$

Objective Function Minimise the total installation cost of all the towers.

$$\text{Min } z = \sum\sum c_{\text{type}} \cdot x_{i,j}$$

2.4 Small Scale Scenario

2.4.1 Demand Requirements

To illustrate all the constraints, we opted to demonstrate a small-scale scenario as illustrated in Figure 6, focusing on the area between Yishun, Ang Mo Kio, Seleta, and Sengkang for calculation. The values for the boxes can be filled in based on our calculations of Demand Satisfaction Constraints in Section 2.3.3. The boxes with no values were omitted; they are non-occupiable areas where a signal is not needed.

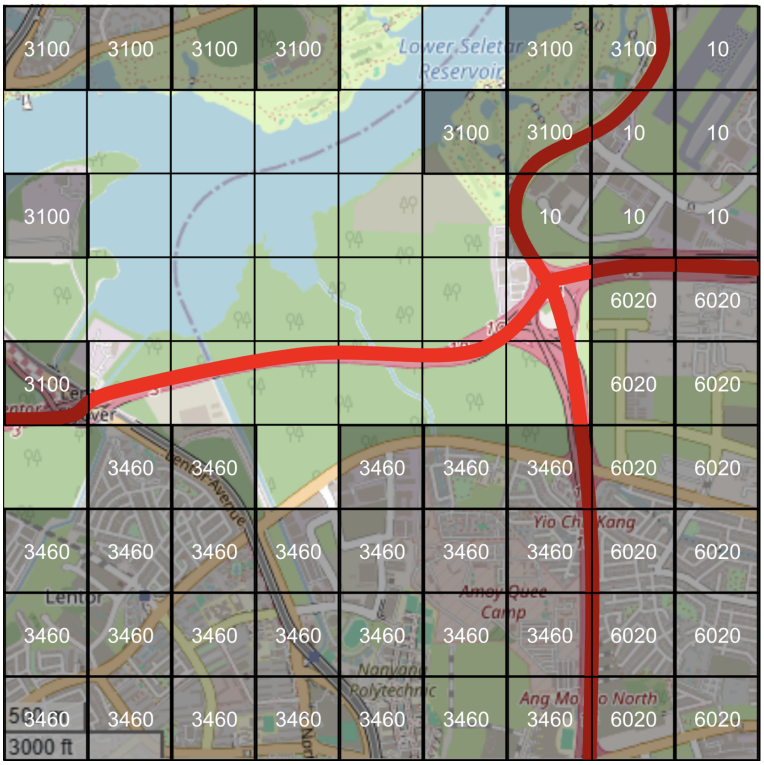


Figure 6: The selected scenario to illustrate all constraints

2.4.2 MIP Solution

In each of the boxes, either all 3 types of towers can be placed or none of them can be placed. The towers share similar constraints to the demand, other than the public space areas. Figure 7 shows our final solution to the MIP for our chosen area in this small scale scenario. Towers can be placed at any of the greyed out areas as depicted in the grid.

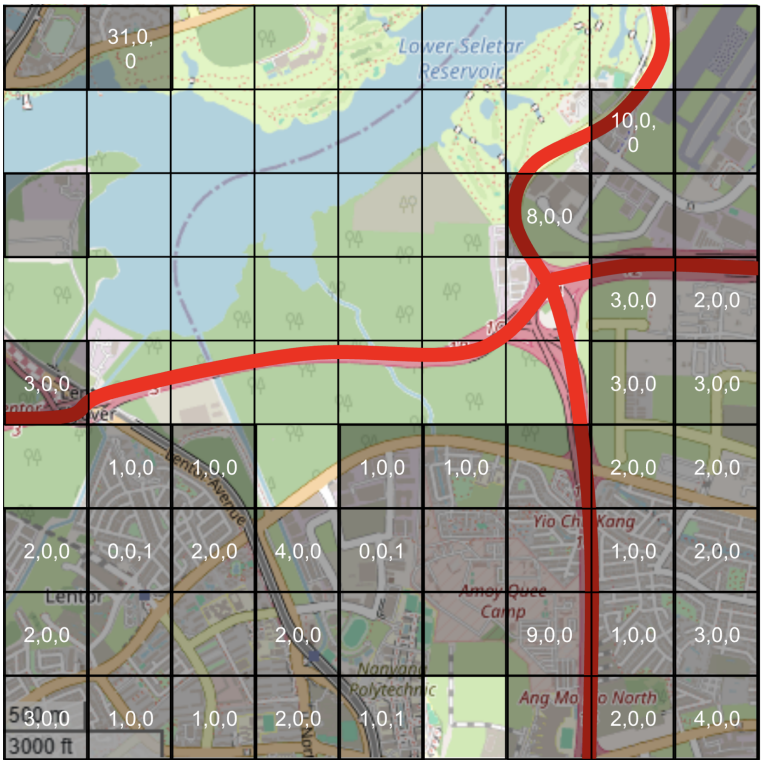


Figure 7: Final Solution to The LP for Our Chosen Area in This Small Scale Scenario. Let (x, y, z) be number of *Micro*, *Small*, and *DSA* tower on each grid. The shaded boxes are the ones we have to ensure sufficient demand for.

In this small-scale scenario, we assumed that no other cell towers are constructed outside the map. Therefore, the towers within the map have to satisfy the demand requirements for the whole map. In the top left corner, the only viable cells that towers can be built on are x_{11} and x_{12}, since cell towers cannot be built on the surrounding terrain without land reclamation. Due to this assumption, our model suggests that 31 micro towers should be built in cell x_{12} to satisfy the local demand. Expanding to a larger map area, taking into consideration land to the north of cell x_{12}, the model could possibly predict a more reasonable spread of cell towers to satisfy the local demand.

3 A More Realistic Scenario to Consider

As people travel during peak hours or attend special events, there may be temporary demand spikes that need to be accounted for. Hence, we introduce a randomized 0–20% increase in the pre-planned demand matrix, as shown in Figure 8.

In addition, we consider the option of hiring land reclamation contractors to reclaim previously inaccessible land at a specified cost, in order to either reduce the total number and cost of signal towers required or ensure that all demand is satisfied. For this scenario, we assume that it would cost \$100,000 to reclaim one land matrix.

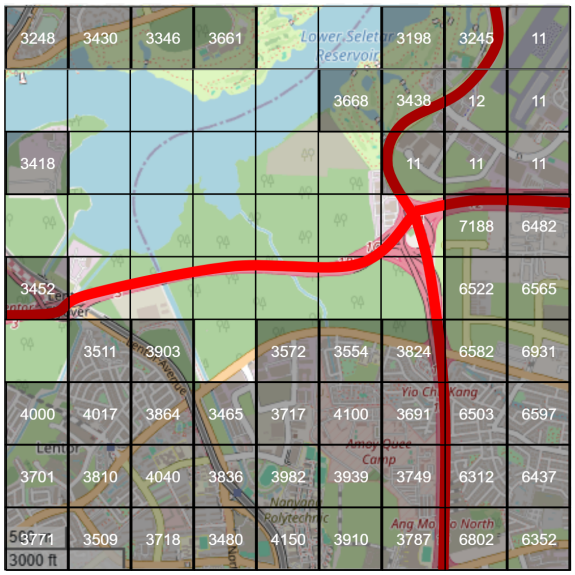


Figure 8: Random Demand

After running the optimisation using Gurobi, we obtain the post-reclamation tower placement shown in Figure 9, where reclaimed land areas are highlighted in red. The model selects these locations as they allow demand to be satisfied with fewer or lower-cost towers overall, demonstrating how land reclamation can influence the cost-effectiveness of the final tower configuration.



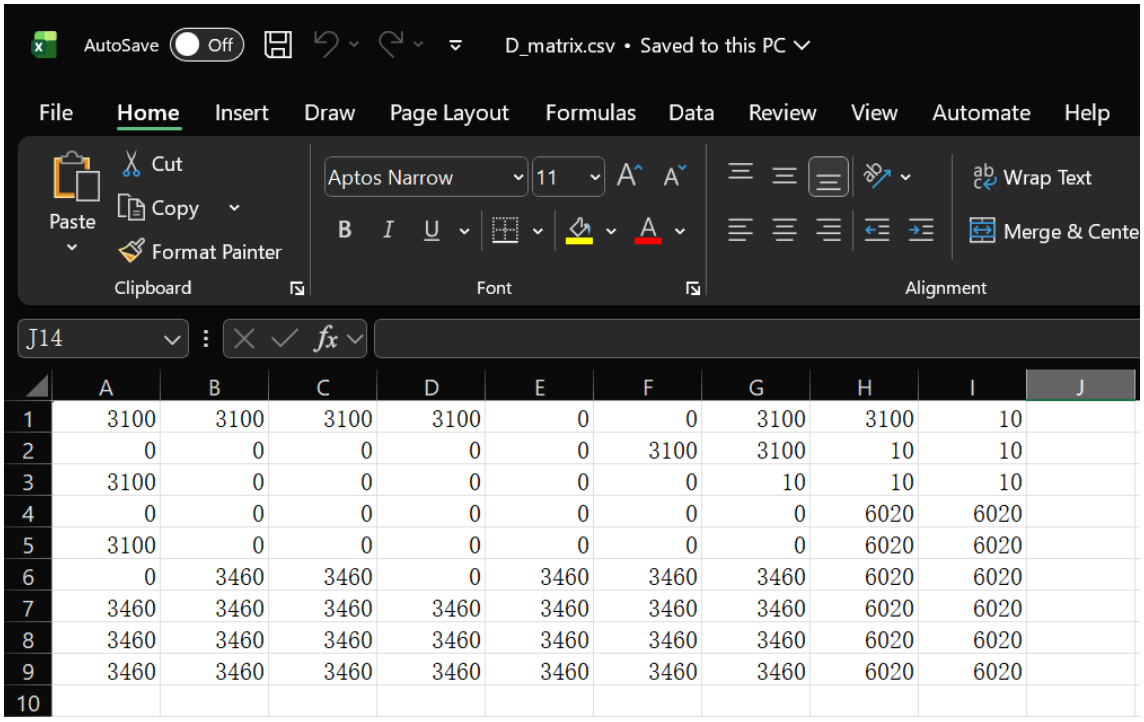
Figure 9: Post-reclamation (in Red) tower placement

This results in a cost of $5,000,000 for tower installation and $300,000 for land reclamation, bringing the total to $5,300,000. Compared to the scenario without reclamation, this configuration achieves full demand coverage while reducing the overall tower deployment cost, illustrating that selective land reclamation can be economically justified.

3.1 Setting a Reasonable Land Reclamation Price

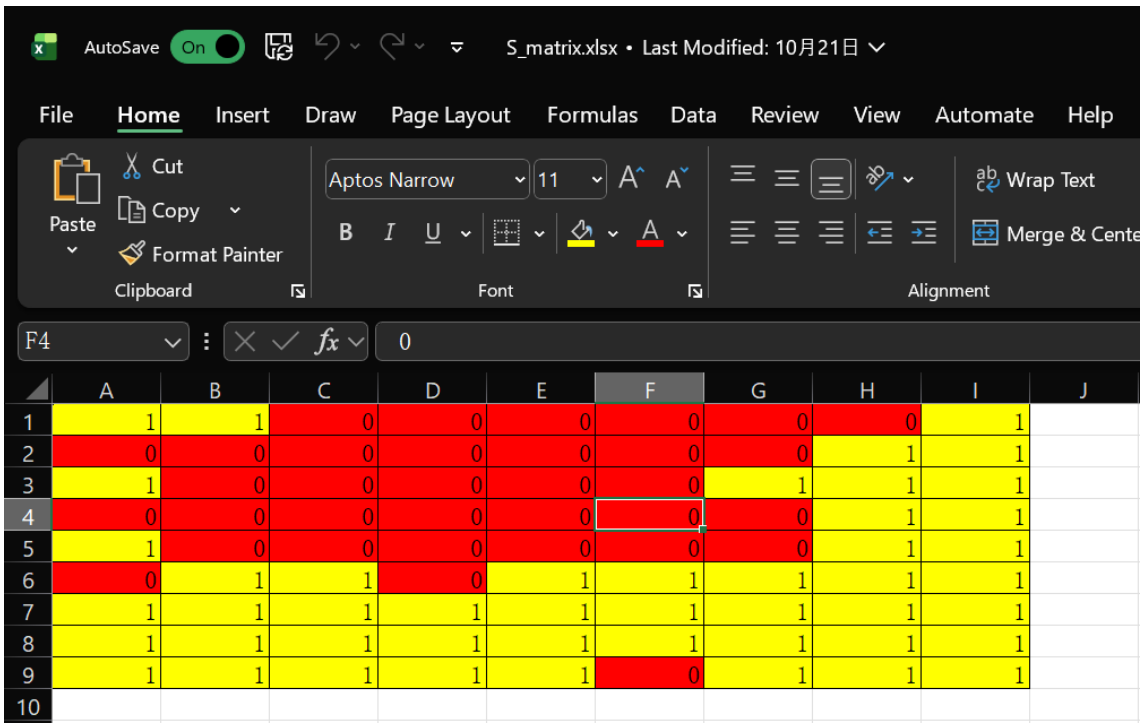
By increasing the cost of land reclamation, we observe that the total installation cost increases from \$4,500,000 to \$5,665,000, beyond which land reclamation is no longer selected by the model. This suggests that above a certain price point, the government would not engage land reclamation contractors to reduce installation cost, unless reclamation is required to meet essential demand.

Therefore, contractors are recommended to set their quoting price below this threshold, which in this case is approximately \$810,000. This value may vary depending on the specific configuration of the $n \times n$ demand matrix (Figure 10), the initial accessible land matrix (Figure 11), the assumed reclamation cost, and the types of signal towers available. Since these parameters can be updated easily through the Excel-based input and output workflow, the model can be readily adapted for different planning scenarios. 13.



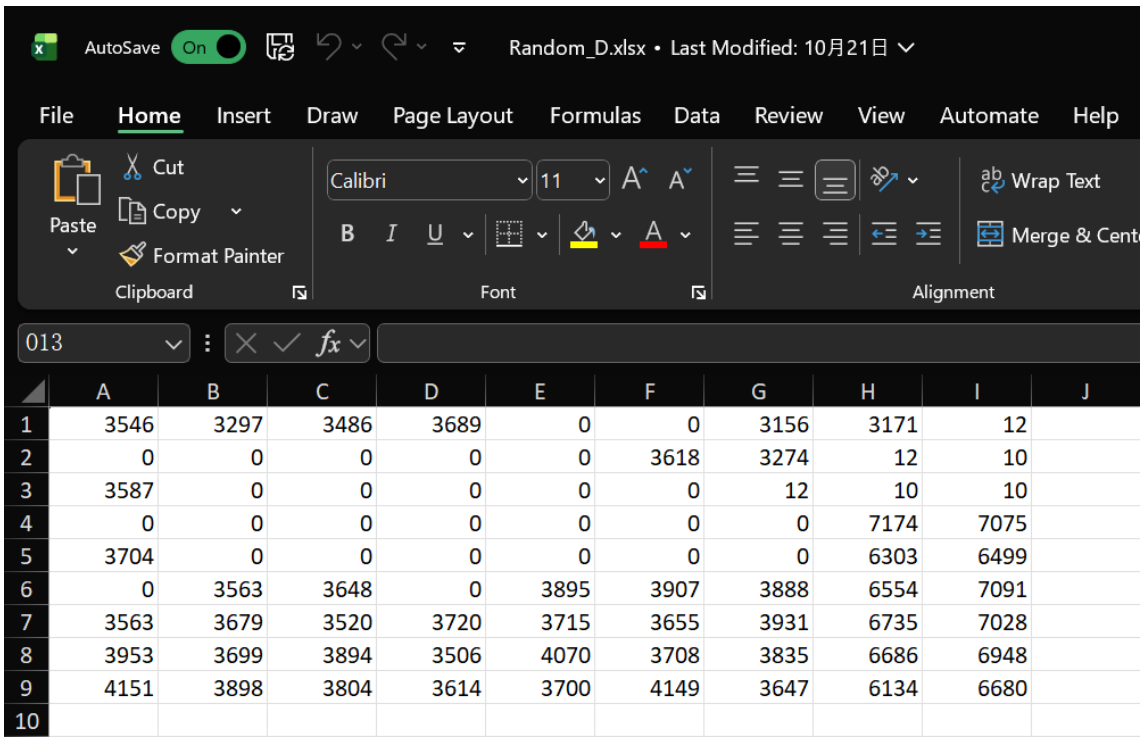
	A	B	C	D	E	F	G	H	I	J
1	3100	3100	3100	3100	0	0	3100	3100	10	
2	0	0	0	0	0	3100	3100	10	10	
3	3100	0	0	0	0	0	10	10	10	
4	0	0	0	0	0	0	0	6020	6020	
5	3100	0	0	0	0	0	0	6020	6020	
6	0	3460	3460	0	3460	3460	3460	6020	6020	
7	3460	3460	3460	3460	3460	3460	3460	6020	6020	
8	3460	3460	3460	3460	3460	3460	3460	6020	6020	
9	3460	3460	3460	3460	3460	3460	3460	6020	6020	
10										

Figure 10: Demand Matrix



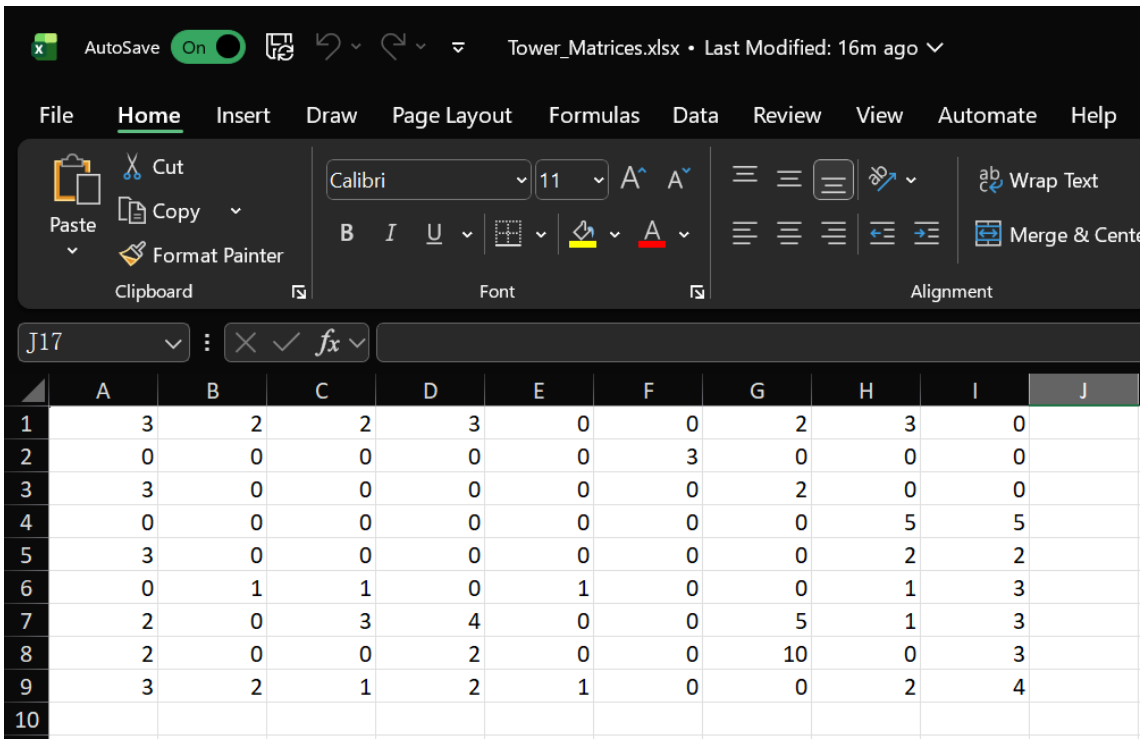
	A	B	C	D	E	F	G	H	I	J
1	1	1	0	0	0	0	0	0	1	
2	0	0	0	0	0	0	0	1	1	
3	1	0	0	0	0	0	1	1	1	
4	0	0	0	0	0	0	0	1	1	
5	1	0	0	0	0	0	0	1	1	
6	0	1	1	0	1	1	1	1	1	
7	1	1	1	1	1	1	1	1	1	
8	1	1	1	1	1	1	1	1	1	
9	1	1	1	1	1	0	1	1	1	
10										

Figure 11: Supply Matrix



	A	B	C	D	E	F	G	H	I	J
1	3546	3297	3486	3689	0	0	3156	3171	12	
2	0	0	0	0	0	3618	3274	12	10	
3	3587	0	0	0	0	0	12	10	10	
4	0	0	0	0	0	0	0	7174	7075	
5	3704	0	0	0	0	0	0	6303	6499	
6	0	3563	3648	0	3895	3907	3888	6554	7091	
7	3563	3679	3520	3720	3715	3655	3931	6735	7028	
8	3953	3699	3894	3506	4070	3708	3835	6686	6948	
9	4151	3898	3804	3614	3700	4149	3647	6134	6680	
10										

Figure 12: Random Demand



	A	B	C	D	E	F	G	H	I	J
1	3	2	2	3	0	0	2	3	0	
2	0	0	0	0	0	3	0	0	0	
3	3	0	0	0	0	0	2	0	0	
4	0	0	0	0	0	0	0	5	5	
5	3	0	0	0	0	0	0	2	2	
6	0	1	1	0	1	0	0	1	3	
7	2	0	3	4	0	0	5	1	3	
8	2	0	0	2	0	0	10	0	3	
9	3	2	1	2	1	0	0	2	4	
10										

Figure 13: Tower Placement

3.2 Future Direction

While the Gurobi model performs sufficiently well for small- and medium-scale scenarios, it may become computationally expensive to obtain optimised solutions when the number of decision variables increases, such as when additional tower types are considered or when the land matrix becomes very large. In such cases, the computation time required for solving the mixed-integer problem may grow significantly.

3.2.1 Branch-Price

To address this, column generation techniques could be incorporated into a branch-and-price framework. Branch-and-price combines column generation with branch-and-bound for enforcing integrality. In this framework, the problem is first formulated as a master problem, and a Restricted Master Problem (RMP) is solved with only a subset of feasible columns. The dual values from the RMP are then used to formulate and solve a subproblem, which identifies new columns (variables) with positive reduced cost to add back into the RMP. This iterative process—solving the RMP, generating duals, solving the subproblem, and adding improving columns—continues until no further improving columns exist [6] as shown in Figure 14. Once the linear relaxation is optimized, branching is performed if the solution is fractional, thereby integrating the column generation procedure within the branch-and-bound tree to enforce integrality [7].

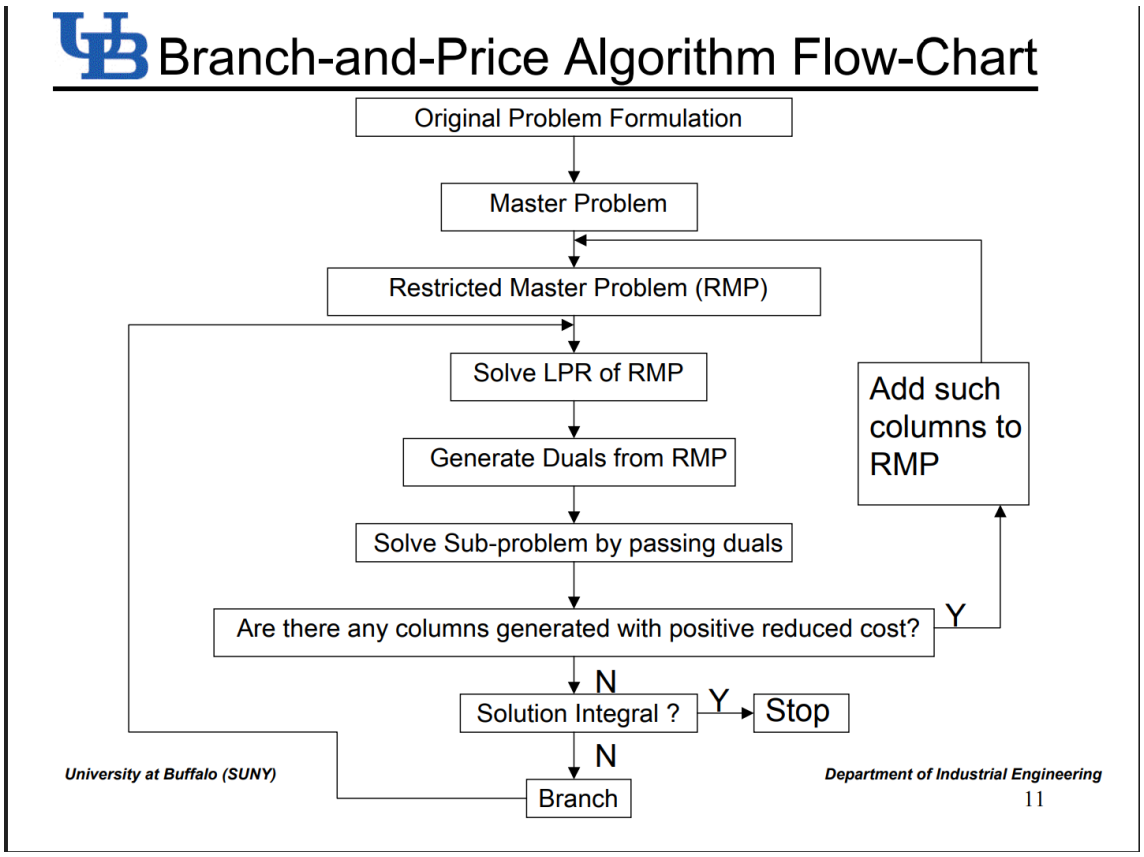


Figure 14: Branch&Price Flowchart

Due to time constraints, these enhancements are not implemented in the present study. However, the current model remains sufficient for the scale of planning scenarios considered here, and future work may explore integrating branch-and-price to improve performance for larger and more complex instances.

4 Conclusion

By modelling Singapore's mobile signal problem as an Operations Research (OR) problem, we were able to apply theoretical OR concepts to a real-life context, generating practical insights that provide tangible value to the various stakeholders involved. This study demonstrates how commerical optimisation tools such as Gurobi can support strategic decision-making in infrastructure planning, particularly when balancing cost and coverage requirements. While the current model focuses on average-scale scenarios, it provides a foundation that can be extended to larger and more complex cases using advanced techniques such as branch-and-price. Overall, this approach illustrates the role of OR in guiding data-driven and cost-effective planning outcomes.

Appendices

Appendix A Python Code to Solve MIP

```
import gurobipy as gp
from gurobipy import GRB
import pandas as pd
import numpy as np


D_df = pd.read_csv("D_matrix.csv", header=None)
S_df = pd.read_excel("S_matrix.xlsx", header=None)

n = D_df.shape[0]
D = D_df.values


D = np.round(D * (1.0 + np.random.uniform(0.0, 0.2, size=D.shape))).astype(int)
with pd.ExcelWriter("Random_D.xlsx", engine="openpyxl") as writer:
    pd.DataFrame(D).to_excel(writer, sheet_name="Random_D", index=False, header=False)
print("Saved uplifted demand matrix to 'Random_D.xlsx'.")


ALL_POS = {(i+1, j+1) for i in range(n) for j in range(n)}

# Land activation cost per site: 0 if S==1, 100000 if S==0
LAND_COST = {(i+1, j+1): (0 if S_df.iat[i, j] == 1 else 100000)
             for i in range(n) for j in range(n)}

# Tower specs
TOWER_SPECS = {
    "x": {"cost": 50000, "strength": {"self": 920,  "N1": 230,  "N2": 101.2}},
    "y": {"cost": 20000, "strength": {"self": 30,   "N1": 7.5}},
    "z": {"cost":  5000, "strength": {"self": 10}},
}
T = list(TOWER_SPECS.keys())

# Neighbour ring helpers
def ring_offsets(ring_name):
    if ring_name == "self":
```

```
        return [(0, 0)]
    if ring_name == "N1":
        return [(di, dj) for di in (-1, 0, 1) for dj in (-1, 0, 1) if not (di == 0 and
    if ring_name == "N2":
        offs = []
        for di in (-2, -1, 0, 1, 2):
            for dj in (-2, -1, 0, 1, 2):
                if (di, dj) != (0, 0) and max(abs(di), abs(dj)) == 2:
                    offs.append((di, dj))
        return list(set(offs))
    return []

RINGS = {"self", "N1", "N2"}
RING_OFFS = {r: ring_offsets(r) for r in RINGS}
def in_grid(i, j, n): return 1 <= i <= n and 1 <= j <= n


m = gp.Model("CoverageWithRings")


VAR_VTYPE = GRB.INTEGER
x = {t: {} for t in T}
for t in T:
    for (i, j) in ALL_POS:
        x[t][(i, j)] = m.addVar(vtype=VAR_VTYPE, lb=0, name=f"{t}{i}{j}")

y = {}
for (i, j) in ALL_POS:
    y[(i, j)] = m.addVar(vtype=GRB.BINARY, name=f"open{i}{j}")

m.update()

# Objective = tower installation costs + land activation costs at S==0 cells
obj_tower = gp.quicksum(
    TOWER_SPECS[t]["cost"] * gp.quicksum(x[t][(i, j)] for (i, j) in ALL_POS)
    for t in T
)
obj_land = gp.quicksum(LAND_COST[(i, j)] * y[(i, j)] for (i, j) in ALL_POS)
m.setObjective(obj_tower + obj_land, GRB.MINIMIZE)
```

```
M = 1000
for (i, j) in ALL_POS:
    m.addConstr(
        gp.quicksum(x[t][(i, j)] for t in T) <= M * y[(i, j)],
        name=f"OpenLink[{i},{j}]"
    )


# Demand coverage constraints
for i in range(1, n+1):
    for j in range(1, n+1):
        lhs_terms = []
        for t in T:
            strengths = TOWER_SPECS[t]["strength"]
            for ring_name, coeff in strengths.items():
                for (di, dj) in RING_OFFS.get(ring_name, []):
                    p, q = i + di, j + dj
                    if in_grid(p, q, n) and (p, q) in x[t]:
                        lhs_terms.append(coeff * x[t][(p, q)])
        m.addConstr(gp.quicksum(lhs_terms) >= D[i-1, j-1], name=f"Demand[{i},{j}]")

m.optimize()


def _build_matrix(n, var_dict):
    Mx = [[0]*n for _ in range(n)]
    for (i, j), var in var_dict.items():
        v = var.X
        if abs(v) > 1e-9:
            Mx[i-1][j-1] = int(v) if abs(v - int(v)) < 1e-9 else float(v)
    return Mx

if m.status == GRB.OPTIMAL:
    print(f"Status: Optimal")
    print(f"Total Installation Cost: ${m.objVal:,.2f}")
    print("\n" + "="*60)

    tower_count = {'x': 0, 'y': 0, 'z': 0}
```

```
    for var in m.getVars():
        if var.x > 0 and var.varName[0] in ('x','y','z'):
            print(f"{var.varName} = {var.x}")
            tower_count[var.varName[0]] += var.x

    print("\n" + "="*60)
    print("SUMMARY STATISTICS:")
    print(f"Total X towers: {tower_count['x']}")
    print(f"Total Y towers: {tower_count['y']}")
    print(f"Total Z towers: {tower_count['z']}")
    print(f"Total all towers: {sum(tower_count.values())}")

    X_mat = _build_matrix(n, x.get("x", {}))
    Y_mat = _build_matrix(n, x.get("y", {}))
    Z_mat = _build_matrix(n, x.get("z", {}))
    out_name = "Tower_Matrices.xlsx"
    with pd.ExcelWriter(out_name, engine="openpyxl") as writer:
        pd.DataFrame(X_mat).to_excel(writer, sheet_name="X_matrix", index=False, heade
        pd.DataFrame(Y_mat).to_excel(writer, sheet_name="Y_matrix", index=False, heade
        pd.DataFrame(Z_mat).to_excel(writer, sheet_name="Z_matrix", index=False, heade

    print(f"Saved X/Y/Z matrices to '{out_name}'.")
else:
    print(f"Status: {m.status}")
    print("No optimal solution found")
```

#1. Implement column generation
#2 Vary the land cost from 0 – a point where land cannot be unlocked.
#3. Save results
#4. Draft paper based on OR1

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